Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

DAIRY FACILITY INFORMATION

A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY: Diamond J Dairy LLC

Physical address of dairy:

Number and Street	City	County	Zip Code
6600 S Hwy 59	Merced	Merced	95341

Street and nearest cross street (if no address):

Date facility was originally placed in operation: 01/01/1993

Regional Water Quality Control Board Basin Plan designation: San Joaquin River Basin

County Assessor Parcel Number(s) for dairy facility:

0066-0220-0013-0000

B. OPERATORS

Jonkman, Wiebren and Luciana

Operator name: Jonkman, Wiebren and Luciana

Telephone no.: (209) 725-2699 (Landline) (Cellular)

Mailing Address Number and Street	City	State	Zip Code
602 E Sandy Mush RD	Merced	CA	95341

This operator is responsible for paying permit fees.

C. OWNERS

Jonkman, Wiebren and Luciana

Legal owner name: Jonkman, Wiebren and Luciana

Telephone no.: (209) 725-2699 (Landline) (Cellular)

Mailing Address Number and Street	City	State	Zip Code
602 E Sandy Mush RD	Merced	CA	95341

This owner is responsible for paying permit fees.

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AVAILABLE NUTRIENTS

A. HERD INFORMATION

	Milk Cows	Dry Cows	Bred Heifers (15-24 mo.)	Heifers (7-14 mo. to breeding)	Calves (4-6 mo.)	Calves (0-3 mo.)
Number open confinement	0	156	200	150	180	0
Number under roof	905	0	50	0	0	0
Maximum number	905	158	250	150	180	0
Average number	905	156	250	150	180	0
Avg live weight (lbs)	1,000	1,050	700	500		

Predominant milk cow breed: Jersey

Average milk production: 67 *pounds per cow per day*

B. MANURE GENERATED

Total manure excreted by the herd: 29,235.56 *tons per reporting period*

Total nitrogen from manure: 374,980.93 *lbs per reporting period*

After ammonia losses (30% loss applied): 262,486.65 *lbs per reporting period*

Total phosphorus from manure: 61,450.39 *lbs per reporting period*

Total potassium from manure: 171,532.10 *lbs per reporting period*

Total salt from manure: 461,991.45 *lbs per reporting period*

C. PROCESS WASTEWATER GENERATED

Process wastewater generated: 22,938,000 *gallons*

Total nitrogen generated: 91,742.66 *lbs*

Total phosphorus generated: 6,945.76 *lbs*

Total potassium generated: 76,663.98 *lbs*

Total salt generated: 960,986.50 *lbs*

	20,618,000	*gallons applied*
+	2,320,000	*gallons exported*
-	0	*gallons imported*
=	22,938,000	*gallons generated*

D. FRESH WATER SOURCES

Source Description	Type
100 Sandy Mush	Ground water
586 Sandy Mush	Ground water
610 Sandy Mush	Ground water
Dairy 1	Ground water
Dairy 2	Ground water

Source Description	Type
Dairy Well East	Ground water
Dairy Well West	Ground water
Front Homes	Ground water
IW #1	Ground water
IW #2	Ground water
IW #3	Ground water
MID	Surface water

E. SUBSURFACE (TILE) DRAINAGE SOURCES

No subsurface (tile) drainage sources entered.

F. NUTRIENT IMPORTS

No dry manure nutrient imports entered.

No process wastewater nutrient imports entered.

No commercial or other nutrient imports entered.

G. NUTRIENT EXPORTS

Date	Material type	Quantity	Reporting basis	Moisture (%)	Density (lbs/cu ft)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/01/2023	Separator solids	557.50 *ton*	Dry-weight	51.8		14,200.00	3,780.00	16,600.00		68.50
11/07/2023	Separator solids	1,850.00 *ton*	Dry-weight	65.0		18,300.00	18,300.00	6,480.00		61.60
11/30/2023	Separator solids	1,125.00 *ton*	Dry-weight	69.8		13,100.00	2,660.00	8,070.00		38.30

Date	Material type	Quantity	Kjeldahl-N (mg/L)	Ammonium-N (mg/L)	Ammonia-N (mg/L)	Nitrate-N (mg/L)	P (mg/L)	K (mg/L)	EC (µmhos/cm)	TDS (mg/L)
09/06/2023	Process wastewater sludge	2,320,000.00 *gal*	1,040.00	126.00	0.00	0.00	48.70	320.00		3,300

Material type	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Dry manure	40,231.46	27,537.46	22,796.50	1,426,108.05
Process wastewater	20,134.82	942.85	6,195.33	63,889.32
Total exports for all materials	60,366.27	28,480.31	28,991.83	1,489,997.37

APPLICATION AREA

A. LIST OF LAND APPLICATION AREAS

Field name	Controlled acres	Cropable acres	Total harvests	Type of waste applied	Parcel number
Field 1	9	9	1	process wastewater	0066-0220-0012-0000
Field 2	12	12	2	process wastewater	0066-0220-0020-0000
Field 3	58	58	2	process wastewater	0066-0220-0020-0000
Field 4	18	18	1	process wastewater	0066-0220-0020-0000
Field 5	50	50	2	process wastewater	0066-0220-0003-0000
Field 6	19	19	2	process wastewater	0066-0220-0020-0000 0066-0220-0021-0000
Field 7	30	30	1	process wastewater	0065-0130-0059-0000
Totals for areas that were used for application	196	196	11		
Totals for areas that were not used for application					
Land application area totals	196	196	11		

B. CROPS AND HARVESTS

Field 1

Field name: Field 1

11/01/2020: Alfalfa, hay

Crop: Alfalfa, hay Acres planted: 9 Plant date: 11/01/2020

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
11/15/2023	81.40 *ton*	As-is		10.0	27,300.00	2,400.00	22,900.00		14.30

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	8.00	480.00	43.20	336.00	0.00
Total actual harvest content	9.04	493.83	43.41	414.24	2,328.04

Field 2

Field name: Field 2

Field 2

09/13/2022: Ryegrass Silage

Crop: Ryegrass Silage Acres planted: 12 Plant date: 09/13/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/13/2023	154.00 *ton*	As-is		56.4	12,560.00	1,870.00	16,667.00		17.40

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	169.60	22.40	224.00	0.00
Total actual harvest content	12.83	322.37	48.00	427.79	1,947.18

06/19/2023: Corn, silage

Crop: Corn, silage Acres planted: 12 Plant date: 06/19/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/29/2023	234.36 *ton*	As-is		72.2	3,450.00	500.00	4,310.00		7.80

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	26.00	208.00	39.00	171.60	0.00
Total actual harvest content	19.53	134.76	19.53	168.35	846.98

Field 3

Field name: Field 3

09/13/2022: Ryegrass Silage

Crop: Ryegrass Silage Acres planted: 58 Plant date: 09/13/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/13/2023	858.40 *ton*	As-is		61.1	10,300.00	1,700.00	15,000.00		18.76

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	169.60	22.40	224.00	0.00
Total actual harvest content	14.80	304.88	50.32	444.00	2,160.10

Field 3

05/16/2023: Corn, silage

Crop: Corn, silage Acres planted: 58 Plant date: 05/16/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
08/25/2023	1,253.90 *ton*	As-is		64.6	4,920.00	780.00	7,550.00		8.58

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	26.00	208.00	39.00	171.60	0.00
Total actual harvest content	21.62	212.73	33.73	326.45	1,313.27

Field 4

Field name: Field 4

11/01/2020: Alfalfa, hay

Crop: Alfalfa, hay Acres planted: 18 Plant date: 11/01/2020

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
11/15/2023	195.00 *ton*	As-is		10.0	28,300.00	2,190.00	25,800.00		15.70

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	8.00	480.00	43.20	336.00	0.00
Total actual harvest content	10.83	613.17	47.45	559.00	3,061.50

Field 5

Field name: Field 5

Field 5

10/10/2022: Wheat, silage, soft dough

Crop: Wheat, silage, soft dough Acres planted: 50 Plant date: 10/10/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/21/2023	559.30 *ton*	As-is		65.8	5,640.00	810.00	7,110.00		15.70

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	176.00	27.20	132.80	0.00
Total actual harvest content	11.19	126.18	18.12	159.06	1,201.24

06/10/2023: Corn, silage

Crop: Corn, silage Acres planted: 50 Plant date: 06/10/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/23/2023	1,415.60 *ton*	As-is		75.8	2,860.00	510.00	5,170.00		8.40

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	26.00	208.00	39.00	171.60	0.00
Total actual harvest content	28.31	161.94	28.88	292.75	1,151.05

Field 6

Field name: Field 6

10/10/2022: Wheat, silage, soft dough

Crop: Wheat, silage, soft dough Acres planted: 19 Plant date: 10/10/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/21/2023	252.20 *ton*	As-is		61.8	6,610.00	1,030.00	7,230.00		20.20

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	176.00	27.20	132.80	0.00
Total actual harvest content	13.27	175.48	27.34	191.94	2,048.50

Field 6

05/26/2023: Corn, silage

Crop: Corn, silage Acres planted: 19 Plant date: 05/26/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/01/2023	369.30 *ton*	As-is		70.8	3,640.00	660.00	6,560.00		9.08

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	26.00	208.00	39.00	171.60	0.00
Total actual harvest content	19.44	141.50	25.66	255.01	1,030.68

Field 7

Field name: Field 7

10/11/2022: Wheat, silage, soft dough

Crop: Wheat, silage, soft dough Acres planted: 30 Plant date: 10/11/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/21/2023	360.10 *ton*	As-is		57.0	6,630.00	1,020.00	5,690.00		9.70

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	176.00	27.20	132.80	0.00
Total actual harvest content	12.00	159.16	24.49	136.60	1,001.32

NUTRIENT BUDGET

A. LAND APPLICATIONS

Field 1 - 11/01/2020: Alfalfa, hay

Field name: Field 1

Crop: Alfalfa, hay

Plant date: 11/01/2020

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
01/02/2023	Surface (irrigation)	Heavy rain	No precipitation	Steady rain

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WW Application	Process wastewater	110.73	12.32	120.70	2,065.11	384,000.00 *gal*
Application event totals		110.73	12.32	120.70	2,065.11	

03/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WW Application	Process wastewater	138.42	15.40	150.88	2,581.39	480,000.00 *gal*
Application event totals		138.42	15.40	150.88	2,581.39	

04/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #1	Ground water	0.00	0.00	0.00	204.29	1,296,000.00 *gal*
Application event totals		0.00	0.00	0.00	204.29	

05/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #1	Ground water	0.00	0.00	0.00	161.73	1,026,000.00 *gal*
Application event totals		0.00	0.00	0.00	161.73	

06/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID	Surface water	0.00	0.00	0.00	123.27	1,564,085.00 *gal*
Application event totals		0.00	0.00	0.00	123.27	

Field 1 - 11/01/2020: Alfalfa, hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID	Surface water	0.00	0.00	0.00	133.54	1,694,425.00 *gal*
Application event totals		0.00	0.00	0.00	133.54	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID	Surface water	0.00	0.00	0.00	130.98	1,661,840.00 *gal*
Application event totals		0.00	0.00	0.00	130.98	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID	Surface water	0.00	0.00	0.00	113.00	1,433,744.00 *gal*
Application event totals		0.00	0.00	0.00	113.00	

Field 2 - 09/13/2022: Ryegrass Silage

Field name: Field 2

Crop: Ryegrass Silage

Plant date: 09/13/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/29/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WW Application	Process wastewater	101.74	7.05	144.47	1,516.79	384,000.00 *gal*
IW #1	Ground water	0.00	0.00	0.00	159.60	1,350,000.00 *gal*
Application event totals		101.74	7.05	144.47	1,676.39	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
01/08/2023	Surface (irrigation)	No precipitation	No precipitation	Light rain

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WW Application	Process wastewater	51.91	5.77	56.58	968.02	240,000.00 *gal*
Application event totals		51.91	5.77	56.58	968.02	

Field 2 - 09/13/2022: Ryegrass Silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
01/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WW Application	Process wastewater	93.43	10.39	101.84	1,742.44	432,000.00 *gal*
Application event totals		93.43	10.39	101.84	1,742.44	

02/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WW Application	Process wastewater	103.81	11.55	113.16	1,936.04	480,000.00 *gal*
Application event totals		103.81	11.55	113.16	1,936.04	

Field 2 - 06/19/2023: Corn, silage

Field name: Field 2

Crop: Corn, silage

Plant date: 06/19/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WW Application	Process wastewater	108.15	35.58	794.44	2,683.75	960,000.00 *gal*
IW #1	Ground water	0.00	0.00	0.00	114.91	972,000.00 *gal*
IW #2	Ground water	42.25	0.00	0.00	653.41	810,000.00 *gal*
Application event totals		150.40	35.58	794.44	3,452.08	

07/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID	Surface water	0.00	0.00	0.00	127.12	2,150,617.00 *gal*
Application event totals		0.00	0.00	0.00	127.12	

Field 2 - 06/19/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID	Surface water	0.00	0.00	0.00	123.27	2,085,446.00 *gal*
Application event totals		0.00	0.00	0.00	123.27	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID	Surface water	0.00	0.00	0.00	115.57	1,955,106.00 *gal*
Application event totals		0.00	0.00	0.00	115.57	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID	Surface water	0.00	0.00	0.00	111.71	1,889,936.00 *gal*
Application event totals		0.00	0.00	0.00	111.71	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID	Surface water	0.00	0.00	0.00	119.42	2,020,276.00 *gal*
Application event totals		0.00	0.00	0.00	119.42	

Field 3 - 09/13/2022: Ryegrass Silage

Field name: Field 3

Crop: Ryegrass Silage

Plant date: 09/13/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/24/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WW Application	Process wastewater	94.73	6.56	134.51	1,412.18	1,728,000.00 *gal*
IW #1	Ground water	0.00	0.00	0.00	167.30	6,840,000.00 *gal*
Application event totals		94.73	6.56	134.51	1,579.48	

Field 3 - 09/13/2022: Ryegrass Silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
01/07/2023	Surface (irrigation)	Light rain	No precipitation	Steady rain

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WW Application	Process wastewater	38.66	4.30	42.14	721.01	864,000.00 *gal*
Application event totals		38.66	4.30	42.14	721.01	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
01/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WW Application	Process wastewater	90.21	10.04	98.33	1,682.35	2,016,000.00 *gal*
Application event totals		90.21	10.04	98.33	1,682.35	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WW Application	Process wastewater	94.50	10.51	103.01	1,762.46	2,112,000.00 *gal*
Application event totals		94.50	10.51	103.01	1,762.46	

Field 3 - 05/16/2023: Corn, silage

Field name: Field 3

Crop: Corn, silage

Plant date: 05/16/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WW Application	Process wastewater	77.32	8.60	84.28	1,442.02	1,728,000.00 *gal*
IW #1	Ground water	0.00	0.00	0.00	126.80	5,184,000.00 *gal*
IW #2	Ground water	30.77	0.00	0.00	475.87	2,851,200.00 *gal*
Application event totals		108.09	8.60	84.28	2,044.68	

Field 3 - 05/16/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID	Surface water	0.00	0.00	0.00	119.55	9,775,530.00 *gal*
Application event totals		0.00	0.00	0.00	119.55	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID	Surface water	0.00	0.00	0.00	115.17	9,417,094.00 *gal*
Application event totals		0.00	0.00	0.00	115.17	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WW Application	Process wastewater	30.94	3.20	43.09	861.89	1,920,000.00 *gal*
MID	Surface water	0.00	0.00	0.00	108.00	8,830,562.00 *gal*
Application event totals		30.94	3.20	43.09	969.89	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WW Application	Process wastewater	110.84	8.10	159.53	1,582.90	1,440,000.00 *gal*
MID	Surface water	0.00	0.00	0.00	111.98	9,156,413.00 *gal*
Application event totals		110.84	8.10	159.53	1,694.88	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID	Surface water	0.00	0.00	0.00	102.42	8,374,371.00 *gal*
Application event totals		0.00	0.00	0.00	102.42	

Field 4 - 11/01/2020: Alfalfa, hay

Field name: Field 4

Crop: Alfalfa, hay

Plant date: 11/01/2020

Field 4 - 11/01/2020: Alfalfa, hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
01/04/2023	Surface (irrigation)	Steady rain	No precipitation	Light rain

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WW Application	Process wastewater	69.21	7.70	75.44	1,290.69	480,000.00 *gal*
Application event totals		69.21	7.70	75.44	1,290.69	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/04/2023	Surface (irrigation)	No precipitation	No precipitation	Light rain

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WW Application	Process wastewater	83.05	9.24	90.53	1,548.83	576,000.00 *gal*
Application event totals		83.05	9.24	90.53	1,548.83	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #1	Ground water	0.00	0.00	0.00	80.86	2,052,000.00 *gal*
Application event totals		0.00	0.00	0.00	80.86	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #1	Ground water	0.00	0.00	0.00	204.29	2,592,000.00 *gal*
Application event totals		0.00	0.00	0.00	204.29	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID	Surface water	0.00	0.00	0.00	125.84	3,193,340.00 *gal*
Application event totals		0.00	0.00	0.00	125.84	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID	Surface water	0.00	0.00	0.00	130.98	3,323,680.00 *gal*
Application event totals		0.00	0.00	0.00	130.98	

Field 4 - 11/01/2020: Alfalfa, hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID	Surface water	0.00	0.00	0.00	120.70	3,062,999.00 *gal*
Application event totals		0.00	0.00	0.00	120.70	

09/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID	Surface water	0.00	0.00	0.00	111.71	2,834,904.00 *gal*
Application event totals		0.00	0.00	0.00	111.71	

Field 5 - 10/10/2022: Wheat, silage, soft dough

Field name: Field 5

Crop: Wheat, silage, soft dough Plant date: 10/10/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/02/2022	Injection	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Slurry Application	Process wastewater sludge	123.24	5.77	37.92	391.05	710,000.00 *gal*
Application event totals		123.24	5.77	37.92	391.05	

01/13/2023	Surface (irrigation)	Light rain	No precipitation	Light rain

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WW Application	Process wastewater	44.85	4.99	48.88	836.37	864,000.00 *gal*
Application event totals		44.85	4.99	48.88	836.37	

Field 5 - 06/10/2023: Corn, silage

Field name: Field 5

Crop: Corn, silage Plant date: 06/10/2023

Field 5 - 06/10/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/25/2023	Injection	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Slurry Application	Process wastewater sludge	166.63	7.80	51.27	528.74	960,000.00 *gal*
Application event totals		166.63	7.80	51.27	528.74	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #1	Ground water	0.00	0.00	0.00	110.31	3,888,000.00 *gal*
IW #2	Ground water	53.62	0.00	0.00	829.40	4,284,000.00 *gal*
Application event totals		53.62	0.00	0.00	939.71	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID	Surface water	0.00	0.00	0.00	126.20	8,895,732.00 *gal*
Application event totals		0.00	0.00	0.00	126.20	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID	Surface water	0.00	0.00	0.00	124.81	8,797,977.00 *gal*
Application event totals		0.00	0.00	0.00	124.81	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID	Surface water	0.00	0.00	0.00	123.43	8,700,222.00 *gal*
Application event totals		0.00	0.00	0.00	123.43	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID	Surface water	0.00	0.00	0.00	118.34	8,341,786.00 *gal*
Application event totals		0.00	0.00	0.00	118.34	

Field 5 - 06/10/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID	Surface water	0.00	0.00	0.00	119.27	8,406,956.00 *gal*
Application event totals		0.00	0.00	0.00	119.27	

Field 6 - 10/10/2022: Wheat, silage, soft dough

Field name: Field 6

Crop: Wheat, silage, soft dough

Plant date: 10/10/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/03/2022	Injection	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Slurry Application	Process wastewater sludge	173.58	8.13	53.41	550.77	380,000.00 *gal*
Application event totals		173.58	8.13	53.41	550.77	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
01/13/2023	Surface (irrigation)	Light rain	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WW Application	Process wastewater	65.57	7.29	71.47	1,222.76	480,000.00 *gal*
Application event totals		65.57	7.29	71.47	1,222.76	

Field 6 - 05/26/2023: Corn, silage

Field name: Field 6

Crop: Corn, silage

Plant date: 05/26/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

Field 6 - 05/26/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/25/2023	Towed tank	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Slurry Application	Process wastewater sludge	132.47	6.20	40.76	420.32	290,000.00 *gal*
Application event totals		132.47	6.20	40.76	420.32	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #1	Ground water	0.00	0.00	0.00	100.80	1,350,000.00 *gal*
IW #2	Ground water	52.57	0.00	0.00	813.14	1,596,000.00 *gal*
Application event totals		52.57	0.00	0.00	913.94	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID	Surface water	0.00	0.00	0.00	122.87	3,291,095.00 *gal*
Application event totals		0.00	0.00	0.00	122.87	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID	Surface water	0.00	0.00	0.00	124.08	3,323,680.00 *gal*
Application event totals		0.00	0.00	0.00	124.08	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID	Surface water	0.00	0.00	0.00	118.00	3,160,755.00 *gal*
Application event totals		0.00	0.00	0.00	118.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID	Surface water	0.00	0.00	0.00	121.65	3,258,510.00 *gal*
Application event totals		0.00	0.00	0.00	121.65	

Field 6 - 05/26/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID	Surface water	0.00	0.00	0.00	119.22	3,193,340.00 *gal*
Application event totals		0.00	0.00	0.00	119.22	

Field 7 - 10/11/2022: Wheat, silage, soft dough

Field name: Field 7

Crop: Wheat, silage, soft dough

Plant date: 10/11/2022

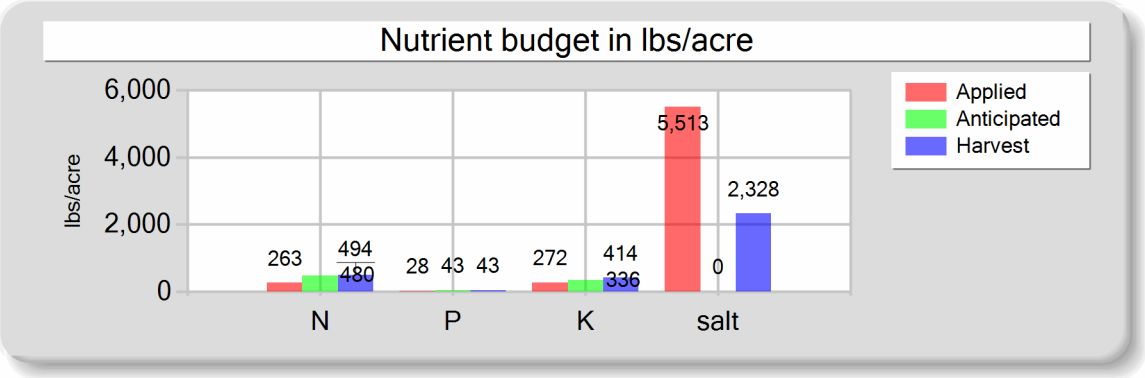
Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/02/2023	Injection	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Slurry Application	Process wastewater sludge	205.40	9.62	63.20	651.74	710,000.00 *gal*
Application event totals		205.40	9.62	63.20	651.74	

B. NUTRIENT BUDGET

Field 1 - 11/01/2020: Alfalfa, hay

Field name: Field 1 Crop: Alfalfa, hay Plant date: 11/01/2020



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	249.15	27.72	271.58	4,646.50
Fresh water	0.00	0.00	0.00	866.80
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	263.15	27.72	271.58	5,513.30
Anticipated crop nutrient removal	480.00	43.20	336.00	0.00
Actual crop nutrient removal	493.83	43.41	414.24	2,328.04
Nutrient balance	-230.68	-15.69	-142.66	3,185.26
Applied to removed ratio	0.53	0.64	0.66	2.37

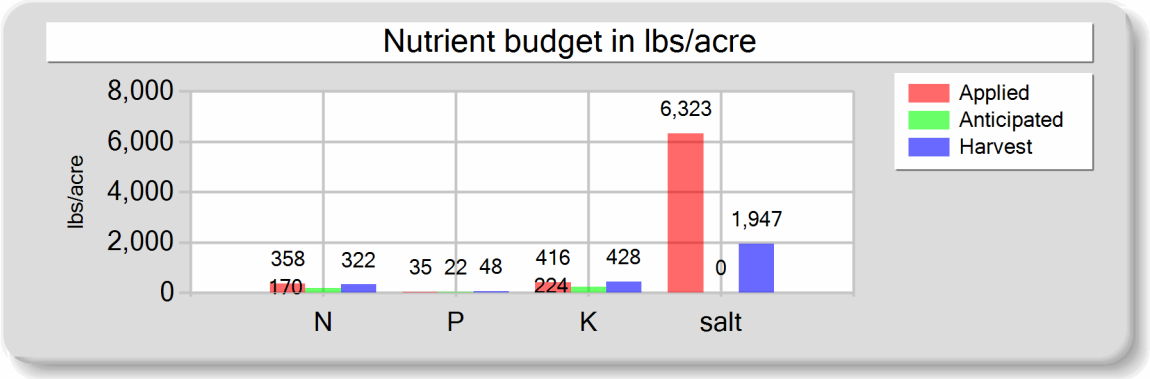
Fresh water applied	
8,676,094.00	*gallons*
319.51	*acre-inches*
35.50	*inches/acre*

Process wastewater applied	
864,000.00	*gallons*
31.82	*acre-inches*
3.54	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 2 - 09/13/2022: Ryegrass Silage

Field name: Field 2 Crop: Ryegrass Silage Plant date: 09/13/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	350.89	34.77	416.05	6,163.28
Fresh water	0.00	0.00	0.00	159.60
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	357.89	34.77	416.05	6,322.88
Anticipated crop nutrient removal	169.60	22.40	224.00	0.00
Actual crop nutrient removal	322.37	48.00	427.79	1,947.18
Nutrient balance	35.52	-13.23	-11.74	4,375.71
Applied to removed ratio	1.11	0.72	0.97	3.25

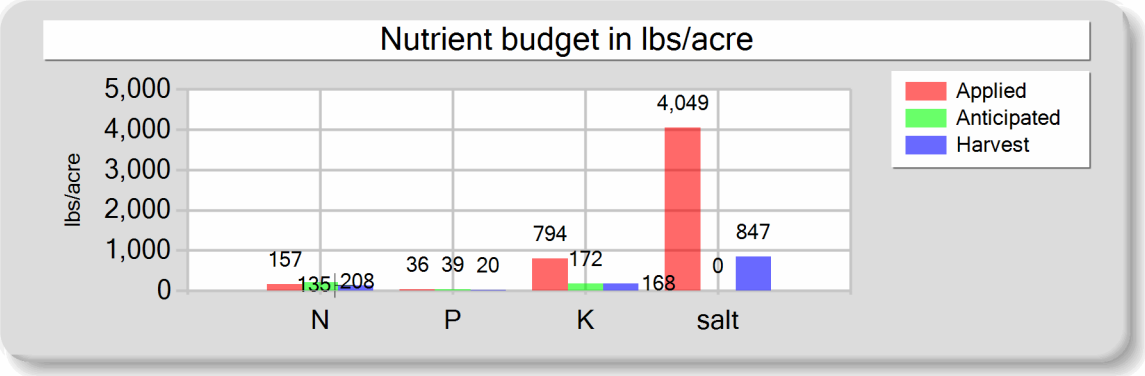
Fresh water applied	
1,350,000.00	*gallons*
49.72	*acre-inches*
4.14	*inches/acre*

Process wastewater applied	
1,536,000.00	*gallons*
56.57	*acre-inches*
4.71	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 2 - 06/19/2023: Corn, silage

Field name: Field 2 Crop: Corn, silage Plant date: 06/19/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	108.15	35.58	794.44	2,683.75
Fresh water	42.25	0.00	0.00	1,365.42
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	157.40	35.58	794.44	4,049.17
Anticipated crop nutrient removal	208.00	39.00	171.60	0.00
Actual crop nutrient removal	134.76	19.53	168.35	846.98
Nutrient balance	22.64	16.05	626.10	3,202.20
Applied to removed ratio	1.17	1.82	4.72	4.78

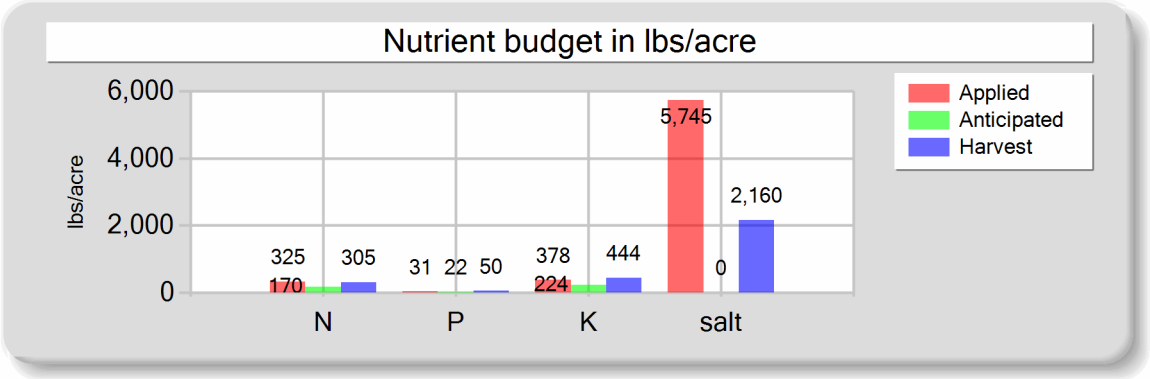
Fresh water applied	
11,883,381.00	*gallons*
437.62	*acre-inches*
36.47	*inches/acre*

Process wastewater applied	
960,000.00	*gallons*
35.35	*acre-inches*
2.95	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 3 - 09/13/2022: Ryegrass Silage

Field name: Field 3 Crop: Ryegrass Silage Plant date: 09/13/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	318.10	31.41	377.99	5,578.01
Fresh water	0.00	0.00	0.00	167.30
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	325.10	31.41	377.99	5,745.31
Anticipated crop nutrient removal	169.60	22.40	224.00	0.00
Actual crop nutrient removal	304.88	50.32	444.00	2,160.10
Nutrient balance	20.22	-18.91	-66.01	3,585.21
Applied to removed ratio	1.07	0.62	0.85	2.66

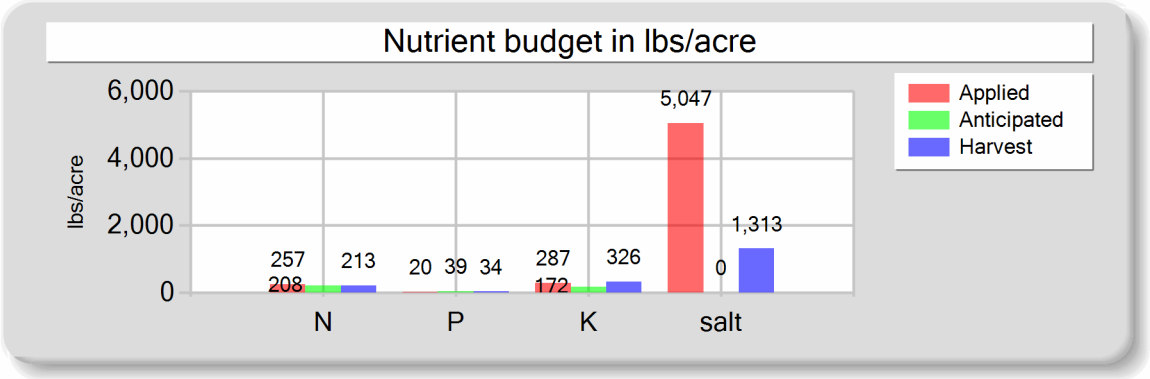
Fresh water applied	
6,840,000.00	*gallons*
251.89	*acre-inches*
4.34	*inches/acre*

Process wastewater applied	
6,720,000.00	*gallons*
247.47	*acre-inches*
4.27	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 3 - 05/16/2023: Corn, silage

Field name: Field 3 Crop: Corn, silage Plant date: 05/16/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	219.11	19.91	286.91	3,886.81
Fresh water	30.77	0.00	0.00	1,159.78
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	256.87	19.91	286.91	5,046.59
Anticipated crop nutrient removal	208.00	39.00	171.60	0.00
Actual crop nutrient removal	212.73	33.73	326.45	1,313.27
Nutrient balance	44.14	-13.82	-39.53	3,733.32
Applied to removed ratio	1.21	0.59	0.88	3.84

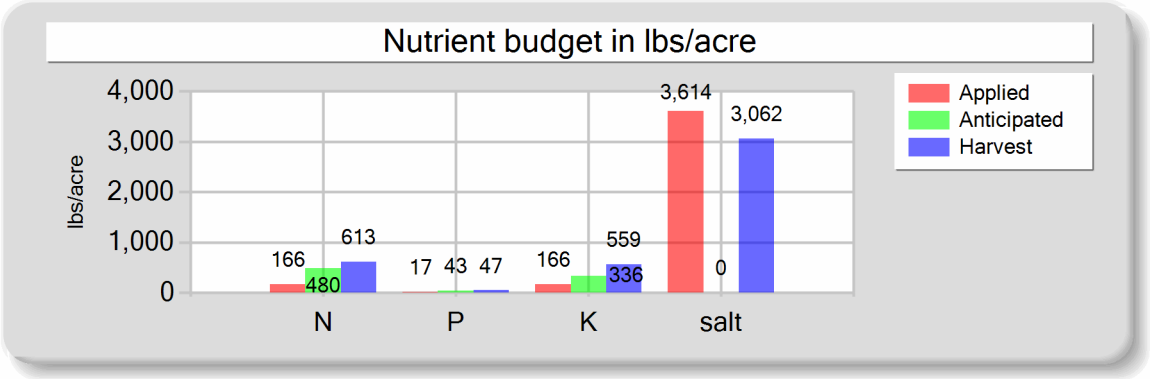
Fresh water applied	
53,589,170.00	*gallons*
1,973.51	*acre-inches*
34.03	*inches/acre*

Process wastewater applied	
5,088,000.00	*gallons*
187.37	*acre-inches*
3.23	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 4 - 11/01/2020: Alfalfa, hay

Field name: Field 4 Crop: Alfalfa, hay Plant date: 11/01/2020



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	152.26	16.94	165.97	2,839.53
Fresh water	0.00	0.00	0.00	774.38
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	166.26	16.94	165.97	3,613.91
Anticipated crop nutrient removal	480.00	43.20	336.00	0.00
Actual crop nutrient removal	613.17	47.45	559.00	3,061.50
Nutrient balance	-446.91	-30.51	-393.03	552.41
Applied to removed ratio	0.27	0.36	0.30	1.18

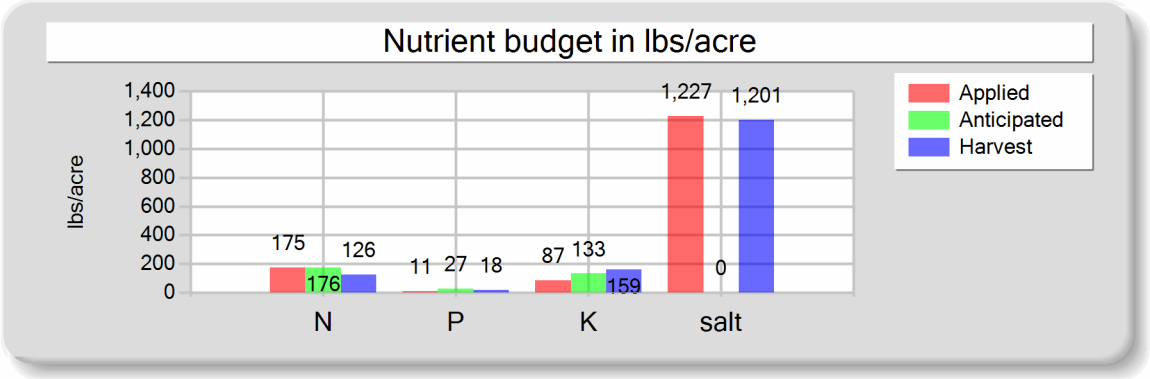
Fresh water applied	
17,058,923.00	*gallons*
628.22	*acre-inches*
34.90	*inches/acre*

Process wastewater applied	
1,056,000.00	*gallons*
38.89	*acre-inches*
2.16	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 5 - 10/10/2022: Wheat, silage, soft dough

Field name: Field 5 Crop: Wheat, silage, soft dough Plant date: 10/10/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	168.09	10.76	86.80	1,227.42
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	175.09	10.76	86.80	1,227.42
Anticipated crop nutrient removal	176.00	27.20	132.80	0.00
Actual crop nutrient removal	126.18	18.12	159.06	1,201.24
Nutrient balance	48.91	-7.36	-72.26	26.17
Applied to removed ratio	1.39	0.59	0.55	1.02

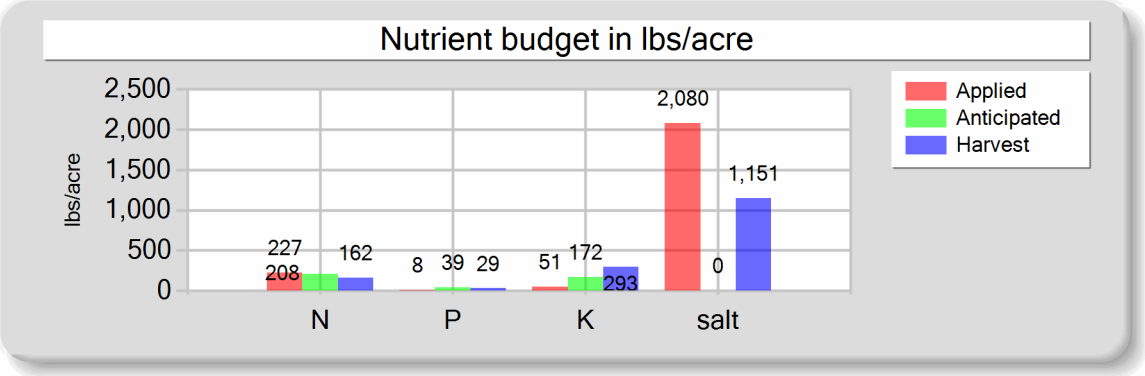
Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
1,574,000.00 *gallons*
57.97 *acre-inches*
1.16 *inches/acre*

Total harvests for the crop
1 *harvests*

Field 5 - 06/10/2023: Corn, silage

Field name: Field 5 Crop: Corn, silage Plant date: 06/10/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	166.63	7.80	51.27	528.74
Fresh water	53.62	0.00	0.00	1,551.76
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	227.26	7.80	51.27	2,080.50
Anticipated crop nutrient removal	208.00	39.00	171.60	0.00
Actual crop nutrient removal	161.94	28.88	292.75	1,151.05
Nutrient balance	65.31	-21.08	-241.47	929.44
Applied to removed ratio	1.40	0.27	0.18	1.81

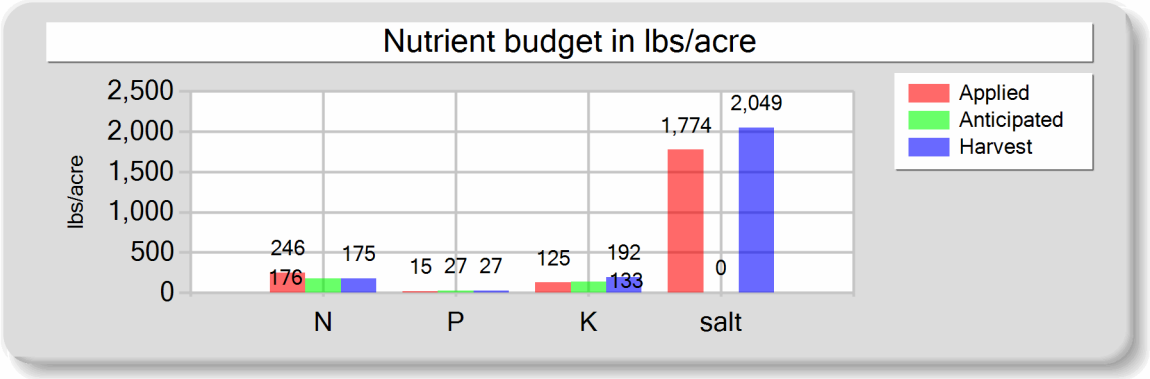
Fresh water applied	
51,314,673.00	*gallons*
1,889.74	*acre-inches*
37.79	*inches/acre*

Process wastewater applied	
960,000.00	*gallons*
35.35	*acre-inches*
0.71	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 6 - 10/10/2022: Wheat, silage, soft dough

Field name: Field 6 Crop: Wheat, silage, soft dough Plant date: 10/10/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	239.14	15.42	124.88	1,773.53
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	246.14	15.42	124.88	1,773.53
Anticipated crop nutrient removal	176.00	27.20	132.80	0.00
Actual crop nutrient removal	175.48	27.34	191.94	2,048.50
Nutrient balance	70.66	-11.92	-67.06	-274.97
Applied to removed ratio	1.40	0.56	0.65	0.87

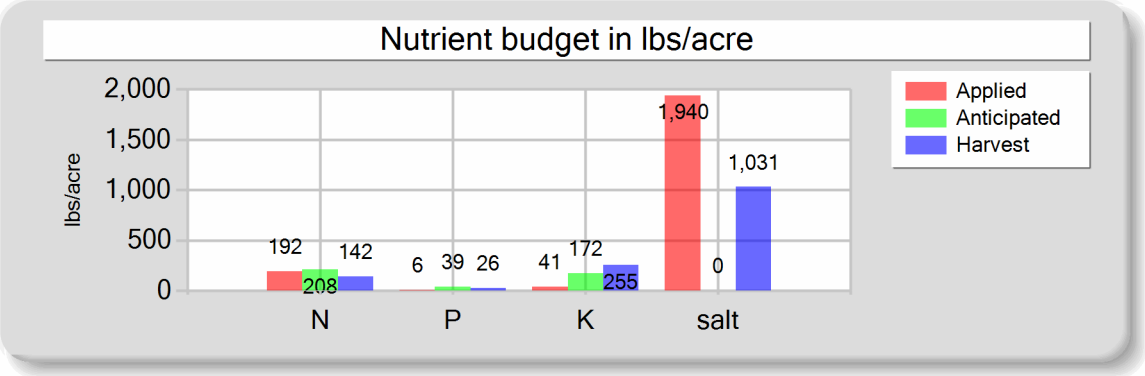
Fresh water applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Process wastewater applied	
860,000.00	*gallons*
31.67	*acre-inches*
1.67	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 6 - 05/26/2023: Corn, silage

Field name: Field 6 Crop: Corn, silage Plant date: 05/26/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	132.47	6.20	40.76	420.32
Fresh water	52.57	0.00	0.00	1,519.75
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	192.04	6.20	40.76	1,940.08
Anticipated crop nutrient removal	208.00	39.00	171.60	0.00
Actual crop nutrient removal	141.50	25.66	255.01	1,030.68
Nutrient balance	50.54	-19.45	-214.25	909.39
Applied to removed ratio	1.36	0.24	0.16	1.88

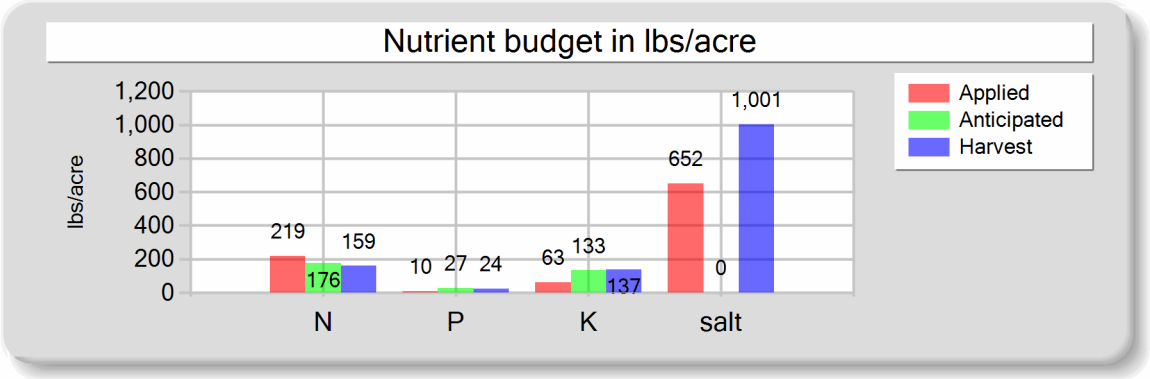
Fresh water applied	
19,173,380.00	*gallons*
706.09	*acre-inches*
37.16	*inches/acre*

Process wastewater applied	
290,000.00	*gallons*
10.68	*acre-inches*
0.56	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 7 - 10/11/2022: Wheat, silage, soft dough

Field name: Field 7 Crop: Wheat, silage, soft dough Plant date: 10/11/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	205.40	9.62	63.20	651.74
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	219.40	9.62	63.20	651.74
Anticipated crop nutrient removal	176.00	27.20	132.80	0.00
Actual crop nutrient removal	159.16	24.49	136.60	1,001.32
Nutrient balance	60.23	-14.87	-73.40	-349.57
Applied to removed ratio	1.38	0.39	0.46	0.65

Fresh water applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Process wastewater applied	
710,000.00	*gallons*
26.15	*acre-inches*
0.87	*inches/acre*

Total harvests for the crop	
1	*harvests*

NUTRIENT ANALYSES

A. MANURE ANALYSES

SM

Sample and source description: SM

Sample date: 12/20/2022 Material type: Separator solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 76.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	17,900.00	3,400.00	13,500.00	15,600.00	7,400.00	3,300.00	5,000.00	1,080.00		21.50
DL	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00		0.01

Spring SM

Sample and source description: Spring SM

Sample date: 05/30/2023 Material type: Separator solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 51.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	14,200.00	3,780.00	16,600.00	19,400.00	7,550.00	3,430.00	3,500.00	1,300.00		68.50
DL	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00		0.01

Export From Lagoon

Sample and source description: Export From Lagoon

Sample date: 09/23/2023 Material type: Separator solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 65.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	18,300.00	18,300.00	6,480.00							61.60
DL	100.00	100.00	100.00							0.01

Fall Exports

Sample and source description: Fall Exports

Sample date: 11/17/2023 Material type: Separator solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 69.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	13,100.00	2,660.00	8,070.00							38.30
DL	100.00	100.00	100.00							0.01

B. PROCESS WASTEWATER ANALYSES

4QWW

Sample and source description: 4QWW

Sample date: 12/20/2022 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	381.00	274.00			26.40	541.00								5,640.00	5,680
DL	0.50	0.50			0.50	0.50								1.00	10

1st QWW

Sample and source description: 1st QWW

Sample date: 03/24/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	311.00	216.00			34.60	339.00								4,530.00	5,800
DL	0.50	0.50			0.50	0.50								1.00	10

2nd QTR WW

Sample and source description: 2nd QTR WW

Sample date: 05/17/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	162.00	118.00			53.30	1,190.00								3,510.00	4,020
DL	0.50	0.50			0.50	0.50								1.00	10

WW

Sample and source description: WW

Sample date: 07/11/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	112.00	81.20			11.60	156.00								2,380.00	3,120
DL	0.50	0.50			0.50	0.50								1.00	10

Slurry

Sample and source description: Slurry

Sample date: 09/07/2023 Material type: Process wastewater sludge Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	1,040.00	126.00			48.70	320.00								5,570.00	3,300
DL	0.50	0.50			0.50	0.50								1.00	10

4th QWW

Sample and source description: 4th QWW

Sample date: 12/27/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	535.00	358.00			39.10	770.00	222.00	123.00	312.00	3,320.00	0.00	233.00	137.00	6,920.00	7,640
DL	0.50	0.50			0.50	0.50	0.50	0.50	0.50	0.50	2.00	0.50	0.50	1.00	10

C. FRESH WATER ANALYSES

100 Sandy Mush

100 Sandy Mush

Sample description: 100 Sandy Mush

Sample date: 12/27/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value		0.00	77.00								2,250.00	
DL		0.50	0.50								1.00	

586 Sandy Mush

586 Sandy Mush

Sample description: 586 Sandy Mush

Sample date: 12/27/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value		0.00	0.00								340.00	
DL		0.50	0.14								1.00	

610 Sandy Mush

610 Sandy Mush

Sample description: 610 Sandy Mush

Sample date: 12/27/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value		0.00	0.00								574.00	
DL		0.50	0.14								1.00	

Dairy Well East

Dairy East

Sample description: Dairy East

Sample date: 12/27/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value		0.00	0.00								403.00	
DL		0.50	0.14								1.00	

Dairy Well West

Dairy West

Sample description: Dairy West

Sample date: 12/27/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value		0.00	0.00								360.00	
DL		0.50	0.14								1.00	

Front Homes

Dairy Houses

Sample description: Dairy Houses

Sample date: 12/27/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value		0.00	15.70								1,030.00	
DL		0.50	0.50								1.00	

IW #1

IW #1

IW #1 - Field #3

Sample description: IW #1 - Field #3

Sample date: 12/27/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.00	0.00	0.00								341.00	170
DL	1.50	0.50	0.14								1.00	10

IW #2

IW 2

Sample description: IW 2

Sample date: 05/17/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	75.00	0.00	72.20								1,910.00	1,160
DL	0.50	0.50	0.50								1.00	10

IW #3

IW #3 - Field #1

Sample description: IW #3 - Field #1

Sample date: 12/27/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	14.60	0.00	14.60								1,050.00	560
DL	0.50	0.50	0.50								1.00	10

MID

MID

Dutchman Creek

Sample description: Dutchman Creek

Sample date: 07/20/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.00										120.00	85
DL	1.00										10.00	10

D. SOIL ANALYSES

No soil analyses entered.

E. PLANT TISSUE ANALYSES

Field 1 - 11/01/2020: Alfalfa, hay

Field 1

Sample and source description: Field 1

Sample date: 10/31/2022 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 42.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	21,400.00	1,990.00	24,300.00		13.10
DL	100.00	100.00	100.00		0.01

Field 1

Sample and source description: Field 1

Sample date: 05/23/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 56.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	15,100.00	1,580.00	14,300.00		12.90
DL	100.00	100.00	100.00		0.01

Field 1 - 11/01/2020: Alfalfa, hay

Field 1

Sample and source description: Field 1

Sample date: 11/01/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 54.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	14,000.00	1,230.00	11,700.00		14.30
DL	100.00	100.00	100.00		0.01

Field 2 - 09/13/2022: Ryegrass Silage

Field 2

Sample and source description: Field 2

Sample date: 11/23/2022 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 56.1 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	17,900.00	2,430.00	22,800.00		23.90
DL	100.00	100.00	100.00		0.01

Field 2 2nd cutting

Sample and source description: Field 2 2nd cutting

Sample date: 02/15/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 48.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	12,700.00	1,850.00	15,600.00		14.00
DL	100.00	100.00	100.00		0.01

Field 2 - 09/13/2022: Ryegrass Silage

Field 2

Sample and source description: Field 2

Sample date: 04/14/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 64.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	7,080.00	1,330.00	11,600.00		14.20
DL	100.00	100.00	100.00		0.01

Field 2 Average

Sample and source description: Field 2 Average

Sample date: 04/14/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 56.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	12,560.00	1,870.00	16,667.00		17.40
DL	100.00	100.00	100.00		0.01

Field 2 - 06/19/2023: Corn, silage

Field 2

Sample and source description: Field 2

Sample date: 09/28/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 72.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,450.00	500.00	4,310.00		7.80
DL	100.00	100.00	100.00		0.01

Field 3 - 09/13/2022: Ryegrass Silage

Field 3 - 09/13/2022: Ryegrass Silage

Field 3

Sample and source description: Field 3

Sample date: 11/23/2022 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 57.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	11,500.00	2,340.00	20,100.00		25.00
DL	100.00	100.00	100.00		0.01

Field 3 2nd cutting

Sample and source description: Field 3 2nd cutting

Sample date: 02/15/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 52.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	13,000.00	1,820.00	16,300.00		17.40
DL	100.00	100.00	100.00		0.01

Field 3

Sample and source description: Field 3

Sample date: 04/14/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 73.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	6,310.00	940.00	8,600.00		13.90
DL	100.00	100.00	100.00		0.01

Field 3 - 09/13/2022: Ryegrass Silage

Field 3 Average

Sample and source description: Field 3 Average

Sample date: 04/14/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 61.1 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	10,300.00	1,700.00	15,000.00		18.80
DL	100.00	100.00	18.80		0.01

Field 3 - 05/16/2023: Corn, silage

Field 3

Sample and source description: Field 3

Sample date: 08/25/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 64.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,920.00	780.00	7,550.00		8.58
DL	100.00	100.00	100.00		0.01

Field 4 - 11/01/2020: Alfalfa, hay

Field 4

Sample and source description: Field 4

Sample date: 10/21/2022 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 58.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	14,800.00	1,730.00	17,700.00		13.30
DL	100.00	100.00	100.00		0.01

Field 4 - 11/01/2020: Alfalfa, hay

Field 4

Sample and source description: Field 4

Sample date: 05/23/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 70.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	10,400.00	1,010.00	9,830.00		12.20
DL	100.00	100.00	100.00		0.01

Field 4

Sample and source description: Field 4

Sample date: 11/01/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 51.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	15,200.00	1,180.00	13,900.00		15.70
DL	100.00	100.00	100.00		0.01

Field 5 - 10/10/2022: Wheat, silage, soft dough

Field 5

Sample and source description: Field 5

Sample date: 04/21/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 65.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,640.00	810.00	7,110.00		15.70
DL	100.00	100.00	100.00		0.01

Field 5 - 06/10/2023: Corn, silage

Field 5 - 06/10/2023: Corn, silage

Field 5

Sample and source description: Field 5

Sample date: 09/23/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 75.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	2,860.00	510.00	5,170.00		8.40
DL	100.00	100.00	100.00		0.01

Field 6 - 10/10/2022: Wheat, silage, soft dough

Field 6

Sample and source description: Field 6

Sample date: 04/21/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 61.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	6,610.00	1,030.00	7,230.00		20.20
DL	100.00	100.00	100.00		0.01

Field 6 - 05/26/2023: Corn, silage

Field 6

Sample and source description: Field 6

Sample date: 09/01/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 70.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,640.00	660.00	6,560.00		9.08
DL	100.00	100.00	100.00		0.01

Field 7 - 10/11/2022: Wheat, silage, soft dough

Field 7

Sample and source description: Field 7

Sample date: 04/21/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 57.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	6,630.00	1,020.00	5,690.00		9.70
DL	100.00	100.00	100.00		0.01

F. SUBSURFACE (TILE) DRAINAGE ANALYSES

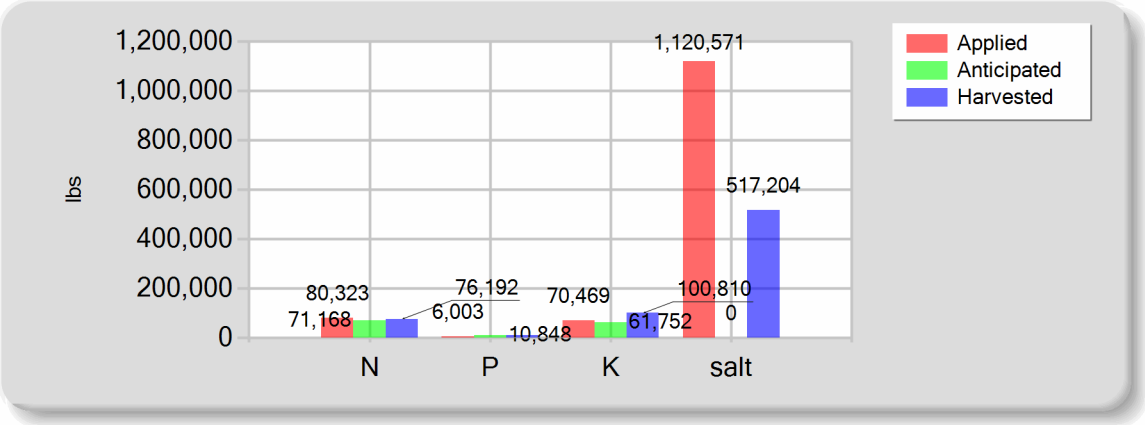
No subsurface (tile) drainage analyses entered.

NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

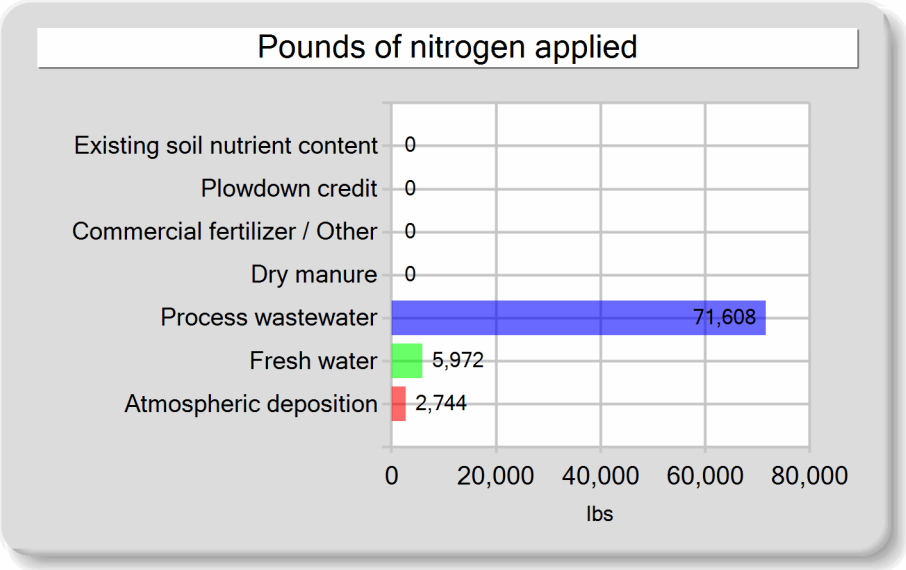
A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

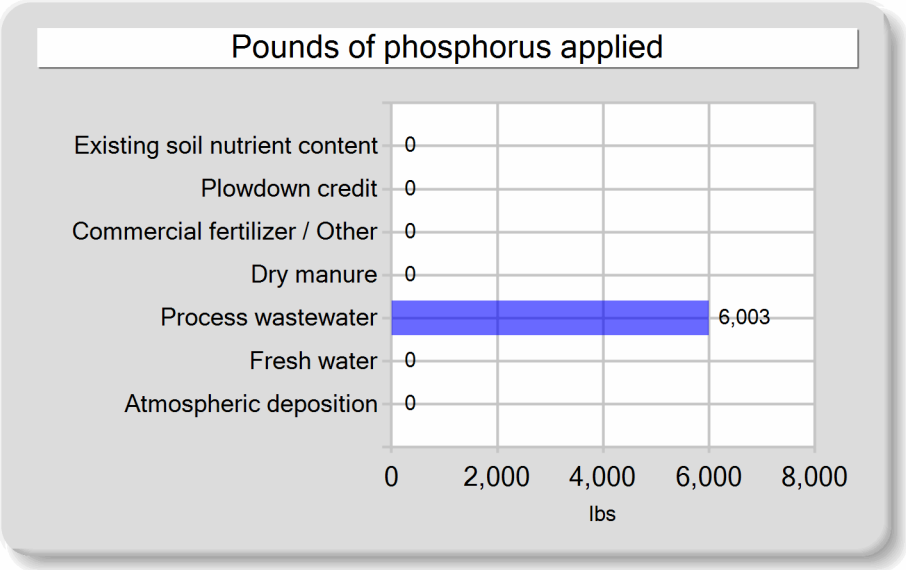
	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	71,607.84	6,002.90	70,468.65	897,097.18
Fresh water	5,971.60	0.00	0.00	223,474.07
Atmospheric deposition	2,744.00	0.00	0.00	0.00
Total nutrients applied	80,323.44	6,002.90	70,468.65	1,120,571.25
Anticipated crop nutrient removal	71,168.00	10,848.20	61,751.60	0.00
Actual crop nutrient removal	76,192.07	11,021.37	100,810.15	517,203.74
Nutrient balance	4,131.37	-5,018.47	-30,341.49	603,367.51
Applied to removed ratio	1.05	0.54	0.70	2.17

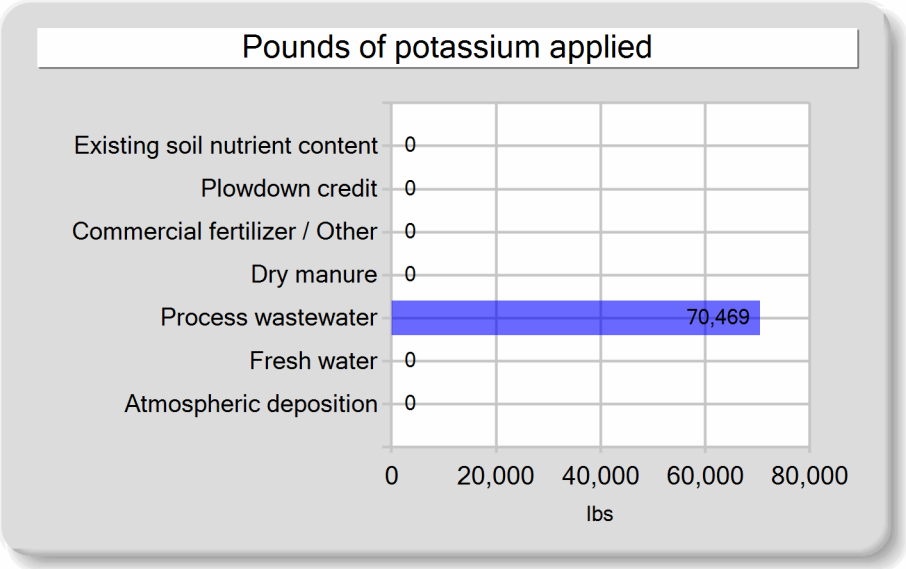
B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL

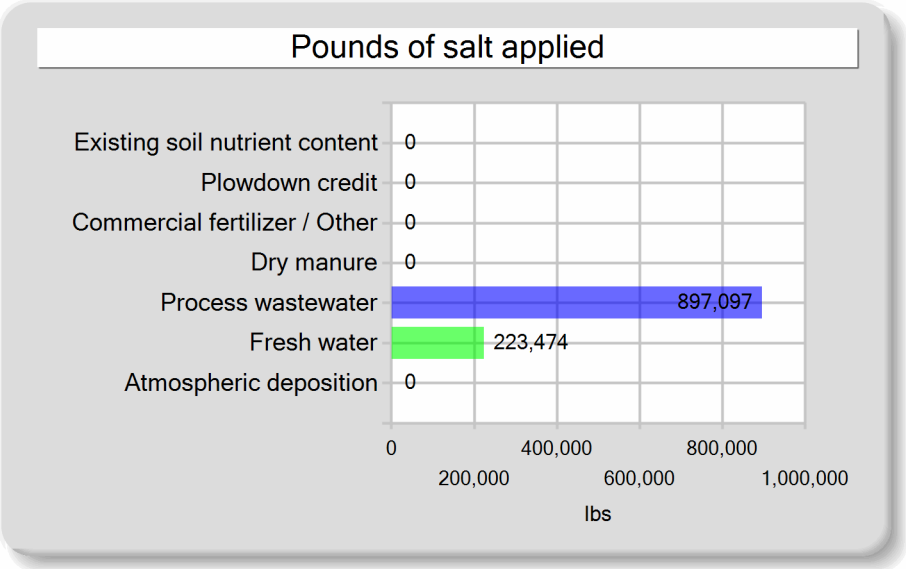


C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE









EXCEPTION REPORTING

A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

No manure or process wastewater discharges occurred during the reporting period.

B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

No stormwater discharges occurred during the reporting period.

C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

No land application area to surface water discharges occurred during the reporting period.

NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

A. NUTRIENT MANAGEMENT PLAN STATEMENTS

Was the facility's NMP updated in the reporting period?	Yes
Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes
Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes

B. EXPORT AGREEMENT STATEMENT

Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period?	No

ADDITIONAL NOTES

A. NOTES

No notes entered for this annual report.

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.

Wiebren Jonkman (Jun 29, 2024 14:20 PDT)	Wiebren Jonkman (Jun 29, 2024 14:20 PDT)
SIGNATURE OF OWNER OF FACILITY	SIGNATURE OF OPERATOR OF FACILITY
Wiebren and Luciana Jonkman	SAME AS OWNER
PRINT OR TYPE NAME	PRINT OR TYPE NAME
06/29/2024	06/29/2024
DATE	DATE

ATTACHMENTS

A. REQUIRED ATTACHMENTS

The following lists the required documents that should be attached to the Annual Report when submitted .

Annual Dairy Facility Assessment
Provide an Annual Dairy Facility Assessment (an update to the Preliminary Dairy Facility Assessment in Attachment A) for each reporting period. On the PDFA Final page, click on the ADFA Report button to generate an ADFA report after updating information as needed .

Manure/Process Wastewater Tracking Manifests
Provide copies of all manure/process wastewater tracking manifests for the reporting period, signed by both the owner/operator and the hauler.

Corrective Actions Documents
Provide records documenting any corrective actions taken to correct deficiencies noted as a result of the inspections required in the Monitoring Requirements of the General Order. Deficiencies not corrected in 30 days must be accompanied by an explanation of the factors preventing immediate correction.

Groundwater Monitoring
Dischargers that monitor supply wells or subsurface (tile) drainage systems, or that have monitoring well systems must submit monitoring results as directed in the General Order, Groundwater Reporting Section starting on page MRP-13.

Storm Water Monitoring
Dischargers that are required to monitor storm water more frequently than required in the General Order must submit monitoring results as directed in the General Order, Storm Water Reporting Section on page MRP-14.

Denele Analytical, Inc.

865 South Kilroy Avenue Turlock, CA 95380
Phone (209) 634-9055 - Fax (209) 634-9057

40 N. East Street, Woodland, CA 95776
Phone (530) 666-9056 - Fax (530) 666-9058

www.denelelabs.com

Compliance Analysis Report

Customer
Diamond J Farms
602 E. Sandy Mush Rd.
Merced, CA 95341

Grower

Consultant:
KR Cordeiro

PURCHASE ORDER:
RECEIVED DATE: 12/28/23 11:10 AM
SUBMITTED BY: KRC
ANALYZED DATE: 1/3/24

Irrigation / Ground Water (H7)
SOURCE: Irrigation Water

NOTES: Temperature was above qualifying at 12.8 C

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
IW #1 - Field #3	T3362017A	Electrical Conductivity (EC)	EPA 120.1 and	0.341	mmhos/cm
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Nitrate Nitrogen (NO3-N)	EPA 353.2	< 0.140	mg/L
		Total Dissolved Solids (TDS)	SM 2540 C-1997 and	170	mg/L
		Total Nitrogen (TN)	SM 4500-NH3 D,C	< 1.50	mg/L
		Soluble Salts (SALT-SOL)	Calculation	218	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	Sample mg/L	Dup mg/L	RPD %
Ammonia Nitrogen (NH3-N)	90.2	97.6 (T3362094B)	99.4 (T3362094B)			1.82
Electrical Conductivity (EC)	95.4			4.00 (T3362091H)	3.98 (Dup)	0.677
Nitrate Nitrogen (NO3-N)	101	92.2 (T3362014C)	91.3 (T3362014C)			1.00
Total Dissolved Solids (TDS)	104			560 (T3362017B)	595 (Dup)	6.06
Total Nitrogen (TN)	97.4			2.80 (T3362014B)	2.80 (null)	0.000

The warranty of Denele Analytical is limited to the accuracy of the analyses of the samples as received. Denele Analytical assumes no responsibility for which the customer uses our test results, nor liability for any other warranties, express or implied. These terms and conditions shall supercede any conflicting terms and conditions submitted on customer purchase orders or other forms submitted for work.

ELAP Certificate No.: 2714

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Compliance Analysis Report

Customer
Diamond J Farms
602 E. Sandy Mush Rd.
Merced, CA 95341

Grower

Consultant:
KR Cordeiro

PURCHASE ORDER:
RECEIVED DATE: 12/28/23 11:10 AM
SUBMITTED BY: KRC
ANALYZED DATE: 1/3/24

Irrigation / Ground Water (H7)
SOURCE: Irrigation Water

NOTES: Temperature was above qualifying at 12.8 C

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
IW #3 - Field #1	T3362017B	Electrical Conductivity (EC)	EPA 120.1 and	1.05	mmhos/cm
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Nitrate Nitrogen (NO3-N)	EPA 353.2	14.6	mg/L
		Total Dissolved Solids (TDS)	SM 2540 C-1997 and	560	mg/L
		Total Nitrogen (TN)	SM 4500-NH3 D,C	14.6	mg/L
		Soluble Salts (SALT-SOL)	Calculation	671	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	Sample mg/L	Dup mg/L	RPD %
Ammonia Nitrogen (NH3-N)	90.2	97.6 (T3362094B)	99.4 (T3362094B)			1.82
Electrical Conductivity (EC)	95.4			4.00 (T3362091H)	3.98 (Dup)	0.677
Nitrate Nitrogen (NO3-N)	101	92.2 (T3362014C)	91.3 (T3362014C)			1.00
Total Dissolved Solids (TDS)	104			560 (T3362017B)	595 (Dup)	6.06
Total Nitrogen (TN)	97.4			2.80 (T3362014B)	2.80 (null)	0.000

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Compliance Analysis Report

Customer
Diamond J Farms
602 E. Sandy Mush Rd.
Merced, CA 95341

Grower

Consultant:
KR Cordeiro

PURCHASE ORDER:
RECEIVED DATE: 12/28/23 11:10 AM
SUBMITTED BY: KRC
ANALYZED DATE: 1/2/24

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 13.0 C

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
100 Sandy Mush	T3362017C	Electrical Conductivity (EC)	EPA 120.1 and	2.25	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	77.0	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	1440	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	91.0			
Electrical Conductivity (EC)	95.4			0.677
Nitrate Nitrogen (NO3-N)	105	99.5 (T3362016J)	98.2 (T3362016J)	1.25

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Compliance Analysis Report

Customer
Diamond J Farms
602 E. Sandy Mush Rd.
Merced, CA 95341

Grower

Consultant:
KR Cordeiro

PURCHASE ORDER:
RECEIVED DATE: 12/28/23 11:10 AM
SUBMITTED BY: KRC
ANALYZED DATE: 1/2/24

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 13.4

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
586 Sandy Mush	T3362017D	Electrical Conductivity (EC)	EPA 120.1 and	0.340	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	< 0.140	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	218	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	91.0			
Electrical Conductivity (EC)	95.4			0.677
Nitrate Nitrogen (NO3-N)	105	99.5 (T3362016J)	98.2 (T3362016J)	1.25

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Grower

Consultant:
KR Cordeiro

PURCHASE ORDER:
RECEIVED DATE: 12/28/23 11:10 AM
SUBMITTED BY: KRC
ANALYZED DATE: 1/2/24

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 14.2

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
610 Sandy Mush	T3362017E	Electrical Conductivity (EC)	EPA 120.1 and	0.574	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	< 0.140	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	367	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	91.0			
Electrical Conductivity (EC)	95.4			0.677
Nitrate Nitrogen (NO3-N)	105	99.5 (T3362016J)	98.2 (T3362016J)	1.25

The warranty of Denele Analytical is limited to the accuracy of the analyses of the samples as received. Denele Analytical assumes no responsibility for which the customer uses our test results, nor liability for any other warranties, express or implied. These terms and conditions shall supercede any conflicting terms and conditions submitted on customer purchase orders or other forms submitted for work.

Denele Analytical, Inc.

865 South Kilroy Avenue Turlock, CA 95380
Phone (209) 634-9055 - Fax (209) 634-9057

40 N. East Street, Woodland, CA 95776
Phone (530) 666-9056 - Fax (530) 666-9058

www.denelelabs.com

Compliance Analysis Report

Customer
Diamond J Farms
602 E. Sandy Mush Rd.
Merced, CA 95341

Grower

Consultant:
KR Cordeiro

PURCHASE ORDER:
RECEIVED DATE: 12/28/23 11:10 AM
SUBMITTED BY: KRC
ANALYZED DATE: 1/2/24

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 14.6

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
Dairy East	T3362017F	Electrical Conductivity (EC)	EPA 120.1 and	0.403	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	< 0.140	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	258	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	91.0			
Electrical Conductivity (EC)	95.4			0.677
Nitrate Nitrogen (NO3-N)	105	99.5 (T3362016J)	98.2 (T3362016J)	1.25

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Compliance Analysis Report

Customer
Diamond J Farms
602 E. Sandy Mush Rd.
Merced, CA 95341

Grower

Consultant:
KR Cordeiro

PURCHASE ORDER:
RECEIVED DATE: 12/28/23 11:10 AM
SUBMITTED BY: KRC
ANALYZED DATE: 1/2/24

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 17.8

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
Dairy West	T3362017G	Electrical Conductivity (EC)	EPA 120.1 and	0.360	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	< 0.140	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	230	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	91.0			
Electrical Conductivity (EC)	95.4			0.677
Nitrate Nitrogen (NO3-N)	105	99.5 (T3362016J)	98.2 (T3362016J)	1.25

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Denele Analytical, Inc.

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www.denelelabs.com

40 N. East Street, Woodland, CA 95776
Phone (530) 666-9056 - Fax (530) 666-9058

Compliance Analysis Report

Customer
Diamond J Farms
602 E. Sandy Mush Rd.
Merced, CA 95341

Grower

Consultant:
KR Cordeiro

PURCHASE ORDER:
RECEIVED DATE: 12/28/23 11:10 AM
SUBMITTED BY: KRC
ANALYZED DATE: 1/2/24

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 16.6

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
Dairy Houses	T3362017H	Electrical Conductivity (EC)	EPA 120.1 and	1.03	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	15.7	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	660	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	91.0			
Electrical Conductivity (EC)	95.4			0.677
Nitrate Nitrogen (NO3-N)	105	99.5 (T3362016J)	98.2 (T3362016J)	1.25

The warranty of Denele Analytical is limited to the accuracy of the analyses of the samples as received. Denele Analytical assumes no responsibility for which the customer uses our test results, nor liability for any other warranties, express or implied. These terms and conditions shall supercede any conflicting terms and conditions submitted on customer purchase orders or other forms submitted for work.

DENELE ANALYTICAL, INC.

865 S. KILROY
TURLOCK, CA 95380

PH# 209-634-9055 FAX# 209-634-9057

DAIRY COMPLIANCE

Client: Diamond J Dairy
Mailing address:
(not facility address) Merced, CA
Phone#:

Authorized Copy Released To: KR Cordeiro
Sampled By: WJ
Project ID:

ANALYSIS TO BE COMPLETED:

IRRIGATION/GROUND WATER
- **H1** EC*, NO3-N, NH4-N*
- **H6** EC, TN, TDS
- **H7** Combination of H1 & H6
- **H14** EC, NO3-N, NH4-N, Ca, Mg, Na, Bicarb Carb, SO4, Cl, TDS

PROCESS WASTE WATER
- **H3** EC*, NH4-N, TKN, TP, K, TDS
- **H4** EC*, NO3-N, NH4-N, TKN, TP, K, TDS
- **H5** Ca, Mg, Na, Bicarb, Carb, SO4, Cl
- **H8** Combination of H3 &H5
- **H9** Combination of H4 &H5

TILE DRAIN
- **H2** EC*, NO3-N, TP, NH4-N*, TDS

** Required Field Measurement*

___ (Check here if performed)

DISCHARGE WATER
- **E1** EC*, Temp*, pH*, NO3-N, NH4-N, TKN, TP, K, TDS, Dissolved Oxygen*, BOD 5, TSS, Total & Fecal Coliforms
- **E2** EC,* Temp*, pH*, NO3-N, NH4-N*, Turbidity, TP, Dissolved Oxygen*, Total & Fecal Coliforms
- **E3** EC*, Temp*, pH*, NO3-N, NH4-N*, TP, Total & Fecal Coliforms
- **E4** EC*, Temp*, pH*, NO3-N, NH4-N*, Turbidity, TP, Total & Fecal Coliform

MANURE
- **M1** % Moisture
- **M3** TN TP, K, % Moisture
- **M4** Ca, Mg, Na, Bicarb, Carb, SO4, Cl, Ash
- **M7** Combination of M3 & M4

PLANT TISSUE
- **F18** % Moisture, Ash, TN, P, K, DM and as received
- **F18+** % Moisture, Ash, TN, P, K, DM and as received (Package also includes NIR DCAD)
- **F19+** % Moisture, TN (Package includes complete plant tissue analysis)

SOIL
- **K3** Soluble Phosphorus
- **K4** 1ft NO3, Organic Matter
- **K5** Pre-Plant (2ft NO3), Fall (1-2ft NO3), Fall (2-3ft NO3)
- **K6** 0-1ft EC, NO3, Soluble P, K, Organic Matter
- **K7** Combination of K3 & K4
- **K8** Combination of K3 & K5

	Date Collected	Time Collected	Sample ID	Description	Temp	Well Purge	Analysis: H1	H3	M3	F18	OTHER
1	12/27/23	12:10 pm	IW #1 - Field 3	IW	12.8						H7
2		12:20 pm	IW #3 - Field 1	IW	12.8						H7
3		12:35 pm	100 Sandy Mush	DW	13.0		X				
4		1:15 pm	586 Sandy Mush	DW	13.4		X				
5		1:12 pm	610 Sandy Mush	DW	14.2		X				
6		12:30 pm	Dairy East	DW	14.6		X				
7		12:31 pm	Dairy West	DW	17.8		X				
8		12:28 pm	Dairy Houses	DW	16.6		X				
9											
10											

CHAIN OF CUSTODY RECORDING

	Signature	Company	Received Date & Time	Relinquished Date & Time	Logged in By:	LABORATORY NUMBER (LABORATORY USE ONLY)
1st	[signature]	KRCordeiro		12/28/23 11:10am		
2nd		DENELE ANALYTICAL, INC.	12-28-23 11:10			T3362017

Required to be submitted to any agency? (YES / NO)

Denele Analytical, Inc.

865 South Kilroy Avenue Turlock, CA 95380
Phone (209) 634-9055 - Fax (209) 634-9057

40 N. East Street, Woodland, CA 95776
Phone (530) 666-9056 - Fax (530) 666-9058

www.denelelabs.com

Compliance Analysis Report

Customer
Diamond J Farms
602 E. Sandy Mush Rd.
Merced, CA 95341

Grower

Consultant:
KR Cordeiro

PURCHASE ORDER:
RECEIVED DATE: 5/19/23 12:05 PM
SUBMITTED BY: KRC
ANALYZED DATE: 5/22/23

Irrigation / Ground Water (H7)
SOURCE: Irrigation Water

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
IW 2	T3139062A	Electrical Conductivity (EC)	EPA 120.1 and	1.91	mmhos/cm
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Nitrate Nitrogen (NO3-N)	EPA 353.2	72.2	mg/L
		Total Dissolved Solids (TDS)	SM 2540 C-1997 and	1160	mg/L
		Total Nitrogen (TN)	SM 4500-NH3 D,C	75.0	mg/L
		Soluble Salts (SALT-SOL)	Calculation	1220	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	Sample mg/L	Dup mg/L	RPD %
Ammonia Nitrogen (NH3-N)	104	100 (T3139044A)	103 (T3139044A)			2.95
Electrical Conductivity (EC)	96.3			3.91 (T3139058A)	3.90 (Dup)	0.256
Nitrate Nitrogen (NO3-N)	107	99.5 (T3142041A)	96.1 (T3142041A)			3.50
Total Dissolved Solids (TDS)	102			245 (null)	235 (Dup)	4.17
Total Nitrogen (TN)	99.1	109 (T3139062A)	109 (T3139062A)			0.000

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ELAP Certificate No.: 2714

DENELE ANALYTICAL, INC.

865 S. KILROY
TURLOCK, CA 95380

PH# 209-634-9055 FAX# 209-634-9057

DAIRY COMPLIANCE

Client: DIAMOND J DAIRY
Mailing address:
(not facility address) MERCED, CA
Phone#:

Authorized Copy Released To: KR CORDEIRO
Sampled By: KRC
Project ID:

ANALYSIS TO BE COMPLETED:

IRRIGATION/GROUND WATER
- H1 EC*, NO3-N, NH4-N*
- H6 EC, TN, TDS
- H7 Combination of H1 & H6
- H14 EC, NO3-N, NH4-N, Ca, Mg, Na, Bicarb Carb, SO4, Cl, TDS

PROCESS WASTE WATER
- H3 EC*, NH4-N, TKN, TP, K, TDS
- H4 EC*, NO3-N, NH4-N, TKN, TP, K, TDS
- H5 Ca, Mg, Na, Bicarb, Carb, SO4, Cl
- H8 Combination of H3 &H5
- H9 Combination of H4 &H5

TILE DRAIN
- H2 EC*, NO3-N, TP, NH4-N*, TDS

** Required Field Measurement*

___ (Check here if performed)

DISCHARGE WATER
- E1 EC*, Temp*, pH*, NO3-N, NH4-N, TKN, TP, K, TDS, Dissolved Oxygen*, BOD 5, TSS, Total & Fecal Coliforms
- E2 EC,* Temp*, pH*, NO3-N, NH4-N*, Turbidity, TP, Dissolved Oxygen*, Total & Fecal Coliforms
- E3 EC*, Temp*, pH*, NO3-N, NH4-N*, TP, Total & Fecal Coliforms
- E4 EC*, Temp*, pH*, NO3-N, NH4-N*, Turbidity, TP, Total & Fecal Coliform

MANURE
- M1 % Moisture
- M3 TN TP, K, % Moisture
- M4 Ca, Mg, Na, Bicarb, Carb, SO4, Cl, Ash
- M7 Combination of M3 & M4

PLANT TISSUE
- F18 % Moisture, Ash, TN, P, K, DM and as received
- F18+ % Moisture, Ash, TN, P, K, DM and as received (Package also includes NIR DCAD)
- F19+ % Moisture, TN (Package includes complete plant tissue analysis)

SOIL
- K3 Soluble Phosphorus
- K4 1ft NO3, Organic Matter
- K5 Pre-Plant (2ft NO3), Fall (1-2ft NO3), Fall (2-3ft NO3)
- K6 0-1ft EC, NO3, Soluble P, K, Organic Matter
- K7 Combination of K3 & K4
- K8 Combination of K3 & K5

	Date Collected	Time Collected	Sample ID	Description	Temp	Well Purge	H1	H3	M3	F18	OTHER
1	5/17/23	2:00pm	IW 2	IW	5.0						H7
2	5/17/23	2:10pm	2nd Q WW	WW	14.0			X			
3	2/15/23		FIELD2 2ND CUTTING	RYE GRASS						X	
4	↓		FIELD3 2ND CUTTING	RYE GRASS						X	
5	4/14/23		FIELD 2	↓						X	
6	↓		FIELD 3	↓						X	
7	4/21/23		FIELD 5	WHEAT						X	
8	↓		FIELD 6	↓						X	
9	↓		FIELD 7	↓						X	
10											

CHAIN OF CUSTODY RECORDING

	Signature	Company	Received Date & Time	Relinquished Date & Time	(LABORATORY USE ONLY) Logged in By:	LABORATORY NUMBER
1st	[illegible]	KR CORDEIRO		5/19/23 12:05pm		
2nd	[illegible]	DENELE ANALYTICAL, INC.	05/19/23 12:05			T3139062

Required to be submitted to any agency? (YES / NO)

Manure / Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

General Order No. R5-2007-0035, Attachment D

INSTRUCTIONS

1) Complete one manifest for each hauling event, for each destination. A hauling event may last for several days, as long as the manure is being hauled to the same destination.
2) If there are multiple destinations, complete a separate form for each destination.
3) The operator must obtain the signature of the hauler upon completion of each manure/process wastewater hauling event.
4) The operator shall submit copies of manure/process wastewater tracking manifest(s) with the Annual Monitoring Report for Existing Milk Cow Dairies.

OPERATOR INFORMATION

Name of Operator: Wiebren and Luciana Jonkman

Name of Dairy Facility: Diamond J Dairy LLC

Facility Address:

Number and Street	City	County	Zip Code
6600 S Hwy 59	Merced	Merced	95341

Contact Person Name and Phone Number: Wiebren and Luciana Jonkman (Name) (209) 725-2699 (Phone Number)

PROCESS WASTEWATER HAULER INFORMATION

Name of Hauling Company/Person: Diamond J Custom Farming, Inc.

Address of Hauling Company/Person:

Number and Street	City	State	Zip Code
602 E Sandy Mush RD	Merced	CA	95341

Contact Person: Wiebren Jonkman (Name) (209) 725-2699 (Phone Number)

DESTINATION INFORMATION

Composting Facility / Broker / Farmer / Other (identify): Farmer

Contact information of Composting Facility, Broker, Farmer, or Other (as identified above):

Name: Red Rock Dairy (Rocky DeJager)
Phone Number: (209) 769-2220

Address	City	State	Zip Code
5020 S 59 HWY	Merced	CA	95340

Destination Address or Assessor's Parcel Number:

Address	City	Zip Code
5020 S Hwy 59	Merced	95340

Street and nearest cross street (if no address)

County: Merced

Assessor's Parcel Number

Assessor's Parcel Number County

Last date hauled: 09/06/2023

Manure / Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

General Order No. R5-2007-0035, Attachment D

PROCESS WASTEWATER AMOUNT HAULED

Enter the amount of process wastewater hauled in gallons and the method used to determine the amount:

Process Wastewater: 2,320,000 gallons

Method used to determine volume of process wastewater:

Tanker Load Volume Multiplied by Total Loads

WRITTEN AGREEMENT

Does the Operator have a written agreement (in compliance with Land Application Specification C.2 of Waste Discharge Requirements General Order No. R5-2007-0035) with any party that receives process wastewater from the Operator for its own use?

[X] YES [ ] NO

If the answer is no, the Operator agrees to have such a written agreement with any such party for any process wastewater transferred after 31 December 2007 to such party.

Wij _____ (Operator shall provide initials here to acknowledge this requirement)

CERTIFICATION

I declare under penalty of law that I personally examined and am familiar with the information submitted in this document, and that based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true , accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of a fine and imprisonment for knowing violations.

Wiebren Jonkman (Jun 29, 2024 14:20 PDT) — 06/29/2024

Operator Signature — Date

Wiebren Jonkman (Jun 29, 2024 14:20 PDT) — 06/29/2024

Hauler Signature — Date

Manure / Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

General Order No. R5-2007-0035, Attachment D

INSTRUCTIONS

1) Complete one manifest for each hauling event, for each destination. A hauling event may last for several days, as long as the manure is being hauled to the same destination.
2) If there are multiple destinations, complete a separate form for each destination.
3) The operator must obtain the signature of the hauler upon completion of each manure/process wastewater hauling event.
4) The operator shall submit copies of manure/process wastewater tracking manifest(s) with the Annual Monitoring Report for Existing Milk Cow Dairies.

OPERATOR INFORMATION

Name of Operator: Wiebren and Luciana Jonkman

Name of Dairy Facility: Diamond J Dairy LLC

Facility Address:

Number and Street	City	County	Zip Code
6600 S Hwy 59	Merced	Merced	95341

Contact Person Name and Phone Number: Wiebren and Luciana Jonkman (Name) (209) 725-2699 (Phone Number)

MANURE HAULER INFORMATION

Name of Hauling Company/Person: Stone Family Spreading Inc.

Address of Hauling Company/Person:

Number and Street	City	State	Zip Code
5175 W Shaw AVE	Winton	CA	95388

Contact Person: Paul Stone (Name) (209) 338-3200 (Phone Number)

DESTINATION INFORMATION

Composting Facility / Broker / Farmer / Other (identify): Farmer

Contact information of Composting Facility, Broker, Farmer, or Other (as identified above):

Ranch Road Farms, LLC (Name) (209) 725-2699 (Phone Number)

Address	City	State	Zip Code
602 E Sandy Mush RD	Merced	CA	95341

Destination Address or Assessor's Parcel Number:

Address	City	Zip Code
1 Ranch RD	Merced	95341

Street and nearest cross street (if no address) — County: Merced

Assessor's Parcel Number — Assessor's Parcel Number County

Last date hauled: 05/01/2023

Manure / Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

General Order No. R5-2007-0035, Attachment D

MANURE AMOUNT HAULED

Enter the amount of manure hauled in tons, manure solids content, and the method used to calculate the amount:

Manure: 557.50 tons

Manure Solids Content: 48.2 %

Method used to determine amount of manure:

Average Load Weight Multiplied by Total Loads

CERTIFICATION

I declare under penalty of law that I personally examined and am familiar with the information submitted in this document, and that based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true , accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of a fine and imprisonment for knowing violations.

Wiebren Jonkman (Jun 29, 2024 14:20 PDT)

Operator Signature — Date: 06/29/2024

Paul Stone

Paul Stone (Jun 26, 2024 13:01 PDT)

Hauler Signature — Date: 06/26/2024

Manure / Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

General Order No. R5-2007-0035, Attachment D

INSTRUCTIONS

1) Complete one manifest for each hauling event, for each destination. A hauling event may last for several days, as long as the manure is being hauled to the same destination.
2) If there are multiple destinations, complete a separate form for each destination.
3) The operator must obtain the signature of the hauler upon completion of each manure/process wastewater hauling event.
4) The operator shall submit copies of manure/process wastewater tracking manifest(s) with the Annual Monitoring Report for Existing Milk Cow Dairies.

OPERATOR INFORMATION

Name of Operator: Wiebren and Luciana Jonkman

Name of Dairy Facility: Diamond J Dairy LLC

Facility Address:

6600 S Hwy 59	Merced	Merced	95341
Number and Street	City	County	Zip Code

Contact Person Name and Phone Number: Wiebren and Luciana Jonkman (209) 725-2699
Name Phone Number

MANURE HAULER INFORMATION

Name of Hauling Company/Person: Stone Family Spreading Inc.

Address of Hauling Company/Person:

5175 W Shaw AVE	Winton	CA	95388
Number and Street	City	State	Zip Code

Contact Person: Paul Stone (209) 338-3200
Name Phone Number

DESTINATION INFORMATION

Composting Facility / Broker / Farmer / Other (identify): Broker

Contact information of Composting Facility, Broker, Farmer, or Other (as identified above):

Stone Family Spreading Inc. (209) 338-3200
Name Phone Number

5175 W Shaw AVE	Winton	CA	95388
Address	City	State	Zip Code

Destination Address or Assessor's Parcel Number:

5175 W Shaw AVE	WInton	95388
Address	City	Zip Code

	Merced
Street and nearest cross street (if no address)	County

Assessor's Parcel Number Assessor's Parcel Number County

Last date hauled: 11/07/2023

Manure / Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

General Order No. R5-2007-0035, Attachment D

MANURE AMOUNT HAULED

Enter the amount of manure hauled in tons, manure solids content, and the method used to calculate the amount:

Manure: 1,850.00 tons

Manure Solids Content: 35.0 %

Method used to determine amount of manure:

Average Load Weight Multiplied by Total Loads

CERTIFICATION

I declare under penalty of law that I personally examined and am familiar with the information submitted in this document, and that based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true , accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of a fine and imprisonment for knowing violations.

Wiebren Jonkman (Jun 29, 2024 14:20 PDT)

Operator Signature — Date: 06/29/2024

Paul Stone

Paul Stone (Jun 26, 2024 13:01 PDT)

Hauler Signature — Date: 06/26/2024

Manure / Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

General Order No. R5-2007-0035, Attachment D

INSTRUCTIONS

1) Complete one manifest for each hauling event, for each destination. A hauling event may last for several days, as long as the manure is being hauled to the same destination.
2) If there are multiple destinations, complete a separate form for each destination.
3) The operator must obtain the signature of the hauler upon completion of each manure/process wastewater hauling event.
4) The operator shall submit copies of manure/process wastewater tracking manifest(s) with the Annual Monitoring Report for Existing Milk Cow Dairies.

OPERATOR INFORMATION

Name of Operator: Wiebren and Luciana Jonkman

Name of Dairy Facility: Diamond J Dairy LLC

Facility Address:

6600 S Hwy 59	Merced	Merced	95341
Number and Street	City	County	Zip Code

Contact Person Name and Phone Number: Wiebren and Luciana Jonkman (209) 725-2699
Name Phone Number

MANURE HAULER INFORMATION

Name of Hauling Company/Person: Stone Family Spreading Inc.

Address of Hauling Company/Person:

5175 W Shaw AVE	Winton	CA	95388
Number and Street	City	State	Zip Code

Contact Person: Paul Stone (209) 338-3200
Name Phone Number

DESTINATION INFORMATION

Composting Facility / Broker / Farmer / Other (identify): Broker

Contact information of Composting Facility, Broker, Farmer, or Other (as identified above):

Stone Family Spreading Inc. (209) 338-3200
Name Phone Number

5175 W Shaw AVE	Winton	CA	95388
Address	City	State	Zip Code

Destination Address or Assessor's Parcel Number:

5175 W Shaw AVE	WInton	95388
Address	City	Zip Code

	Merced
Street and nearest cross street (if no address)	County

Assessor's Parcel Number Assessor's Parcel Number County

Last date hauled: 11/30/2023

Manure / Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

General Order No. R5-2007-0035, Attachment D

MANURE AMOUNT HAULED

Enter the amount of manure hauled in tons, manure solids content, and the method used to calculate the amount:

Manure: 1,125.00 tons

Manure Solids Content: 30.2 %

Method used to determine amount of manure:

Average Load Weight Multiplied by Total Loads

CERTIFICATION

I declare under penalty of law that I personally examined and am familiar with the information submitted in this document, and that based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true , accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of a fine and imprisonment for knowing violations.

Wiebren Jonkman (Jun 29, 2024 14:20 PDT)

Operator Signature — Date: 06/29/2024

Paul Stone

Paul Stone (Jun 26, 2024 13:01 PDT)

Hauler Signature — Date: 06/26/2024